Strategic Plan

Amateur Radio Satellites and System Canada

May 18, 2026

Table of Contents

Glossary of Key Terms

AMSAT - Radio Amateur Satellite Corporation

AMSAT-CA - Amateur Radio Satellites and Systems Canada

AMSAT-DL - German Amateur Radio Satellite Organization

AMSAT-UK - United Kingdom Amateur Radio Satellite Organization

AO-x - AMSAT OSCAR x (x is numeric designator)

APRS - Automatic Packet Reporting System

ARISS - Amateur Radio on the International Space Station

BoD - Board of Directors

DSLWP-A1 - Discovering the Sky at Longest Wavelengths Pathfinder A1

DSLWP-A2 - Discovering the Sky at Longest Wavelengths Pathfinder A2

DVB-S - Digital Video Broadcasting - Satellite

DVB-S2 - Digital Video Broadcasting - Satellite - Second Generation

DVB-S2X - Digital Video Broadcasting - Satellite - Second Generation Extension

DX - Long-distance radio communication

FM - Frequency Modulation

GEO - Geostationary Earth Orbit

HAB - High-Altitude Balloon

HEO - Highly Elliptical Orbit

LEAP - Lunar Exploration Accelerator Program

LEO - Low Earth Orbit

LoRa - Long Range (low-power wide-area radio technology)

MMIC - Monolithic Microwave Integrated Circuit

MINERAL - Multispectral Imager with Navigational Exploration Relay and Autonomous Lander

OSCAR - Orbiting Satellite Carrying Amateur Radio

PLEO - Proliferated Low Earth Orbit

R&D - Research and Development

RAC - Radio Amateurs of Canada

RF - Radio Frequency

SDR - Software-Defined Radio

SLIM - Smart Lander for Investigating Moon

STEM - Science, Technology, Engineering and Mathematics

SWaP - Size, Weight and Power

TT&C - Telemetry, Tracking and Command

UHF - Ultra High Frequency

US - United States

VHF - Very High Frequency

WOC - Weighted Objectives Chart

Letter from the President

May 2026

Fellow AMSAT-CA Members, Partners, and Friends,

It is my privilege to present the first Strategic Plan of Amateur Radio Satellites and Systems Canada (AMSAT-CA). Since our incorporation in 2023, our organization has grown from an idea shared among a handful of dedicated Canadian Amateur satellite enthusiasts into a national entity recognized by both Radio Amateurs of Canada and the international AMSAT community. This document marks an important milestone in our development: the articulation of a clear technical vision and roadmap that will guide our efforts through the end of the decade, and beyond.

Canada has a rich history of innovation in space and radio science, and AMSAT-CA stands ready to contribute to that legacy. Our mandate-to promote, develop, and support Amateur radio in space-reflects the long tradition of experimentation, education, and international collaboration that defines our service. However, as the global space environment becomes increasingly dynamic, competitive, and commercially driven, it is essential that we focus our limited volunteer resources on initiatives where we can have the greatest national and international impact.

This Strategic Plan identifies two complementary pillars on which AMSAT-CA will build its technical identity. First, we aim to become a world leader in accessible, beginner-focused resources and technologies-lowering barriers to participation through open-source OSCAR ground-station kits and high-altitude balloon platforms that welcome students, educators, and new Amateurs into the satellite domain. Second, we aspire to develop Canada's first contributions to lunar and cislunar Amateur-radio experimentation through simple, robust, one-way hosted payloads that demonstrate our capabilities and place Canadian Amateur radio at the forefront of deep-space citizen science.

These priorities are not merely technical choices; they are strategic expressions of who we are as a community. They emphasize innovation, feasibility, education, collaboration, and a uniquely Canadian contribution to the global Amateur-satellite landscape. Most importantly, they provide rallying points-projects that our members, universities, youth groups, and partner organizations can unite behind.

The path ahead will require commitment, creativity, and a shared sense of purpose. As a volunteer organization, our greatest strength is the expertise and enthusiasm of our members. I invite each of you to participate in shaping the future of Amateur radio in space-whether by contributing to design work, supporting outreach, helping secure partnerships, or simply spreading the word about AMSAT-CA's mission.

Together, we can build a vibrant and enduring Canadian presence in the Amateur-satellite service-one that inspires new generations, advances open science, and upholds the spirit of exploration that has always defined our hobby.

Thank you for your continued dedication and support.

Levente Buzas Digitally signed by Levente Buzas Date: 2026.07.18 13:33:51 -07'00'

Levente Buzas, VA7QF
President, AMSAT-CA

Introduction

AMSAT-CA Background

Amateur Radio Satellites and Systems Canada (AMSAT-CA) was incorporated as a federal not for profit corporation on November 12, 2023 by Dr. Stefan Wagner VE4SW and six other Canadian Amateur satellite operators and space industry professionals to provide a venue for Canadian Amateur satellite enthusiasts to exchange ideas and collaborate, and to connect Canada to the highly successful international community of AMSAT organizations.

After its establishment, the organization became recognized by the Radio Amateurs of Canada (RAC) as Canada's official AMSAT organization in May 2024, in the form of a Memorandum of Understanding (MoU) signed between the two organizations [1]. This was followed by international recognition in August 2024, where the original AMSAT organization from the United States granted the organization the right to use the AMSAT name, recognizing it as Canada's official member of the AMSAT family [2].

With its organizational structure and business processes established, including the adaptation of the first set of Bylaws [3] for the organization in May 2025; AMSAT-CA is now looking to define its technical development identity. This official Strategic Plan is a key element of this process, and it intends to establish and present the organization's vision in terms of the concrete technical projects it intends to prioritize.

AMSAT-CA Purpose

The core purpose AMSAT-CA, as defined in its Certificate of Incorporation, is to promote, develop, and support Amateur radio in space, including but not limited to Amateur radio communication via human-made and natural satellites (e.g., the Moon), Amateur radio on space stations and planets, and related experiments, as well as new developments of (i) technology, (ii) methods and techniques, and (iii) new applications, all intended for use on satellites, ground stations, and terrestrial-space communications for the benefit of humanity.

To this end, the key objectives of AMSAT-CA are as follows:

1. AMSAT-CA will promote, develop, and support the education, training, and skills of Amateur radio operators and those interested in Amateur radio in space (e.g., STEM education).
2. AMSAT-CA will promote collaboration with the Canadian universities and colleges involved in the development of satellites and systems.
3. AMSAT-CA will promote, develop, and support innovation in science, engineering and technology, as well as its open access under the principles of "Open Source" and "Open Access to Research".
4. AMSAT-CA will promote and assist in emergency and disaster preparedness.

5. AMSAT-CA will act as a liaison for Amateur radio in space for Canadian Amateur radio associations, societies, organizations, and its members.
6. AMSAT-CA will act as a liaison and consultative entity to municipal, provincial, and federal governments in matters concerning Amateur radio in space and as a liaison and consultative entity to the Canadian government and its departments, while respecting the role of the RAC in this process, as defined in the MoU between the two organizations.
7. AMSAT-CA will promote and support national and international cooperation and Amateur radio satellite organizations for the common good and peaceful exploration of space.

The Need for a Vision for Canada

A well-defined technical vision is essential for guiding AMSAT-CA through its critical formative years leading to the 2030s. This vision and associated Strategic Plan is being prepared by a Working Group designated and authorized by the Board of Directors for this purpose during its regular board meeting held on February 10, 2025. This working group consists of the following individuals:

- Chair - Levente Buzas, VA7QF, President
- Group Member - Scott Tilley, VE7TIL, Vice-President
- Group Member - Brent Betersen, VE9EX, Technical Director

As a national organization rooted in volunteerism, collaboration, and innovation, AMSAT-CA must provide clear direction to align the efforts of engineers, scientists, educators, and Amateur radio operators across Canada. A shared technical roadmap not only ensures coherence and efficiency in project development, but it also helps build momentum, foster community engagement, and attract new talent-particularly among students and early-career professionals in STEM fields, or those experienced folks who, lacking a domestic alternative, were historically members of other AMSAT organizations.

In the rapidly evolving landscape of space technology, a strong technical vision allows AMSAT-CA to anticipate trends, seize opportunities, and focus its limited resources on projects with the highest impact. It ensures that Canadian contributions to the global AMSAT community are both meaningful and forward-looking, enabling homegrown innovation in satellite communications, space science, and open-source technologies. Furthermore, most importantly, a strong technical vision provides a rallying point for all stakeholders - a clear articulation of where we're going, why it matters, and how each person can contribute to making Canada a leader in Amateur radio in space.

The rest of the document has been prepared and organized to establish this technical vision for AMSAT-CA. First, a brief overview of the history of Amateur-satellite activities is presented, to give context to the subsequent review of the state of the art in terms of technical activities which may be suitable for an AMSAT organization. Then, a quantitative criteria is established and applied to evaluate a list of concrete potential projects deemed in line with the Purpose of

AMSAT-CA stated earlier in this document; defined by the technical experts of the AMSAT-CA Working Group responsible for the Strategic Plan. Finally, the outcome of this evaluation is presented and discussed, establishing and presenting the much-needed strategic vision for Amateur Radio Satellites and Systems Canada.

The History of the Amateur Satellite Service

Overview

The history of Amateur satellites is deeply intertwined with the broader story of space exploration, innovation, and international cooperation. It is a history defined not by government budgets or commercial interests, but by passionate individuals-engineers, students, and Radio Amateurs - who saw space not as a distant frontier, but as a place where anyone could contribute and communicate.

The Dawn of Amateur Satellite

The community's Amateur satellite journey began in 1961 with the launch of OSCAR 1, just four years after the Soviet Union's Sputnik became the world's first artificial satellite [4]. OSCAR 1 was built by a small team of Amateur radio operators and engineers in California under the newly formed Project OSCAR group. Weighing just over 10 pounds, and operating on 144.983 MHz, OSCAR 1 transmitted a simple beacon signal ("HI") and remained active for 22 days. Its launch made history-it was the first privately constructed, non-governmental satellite to be placed in orbit.

Following the success of OSCAR 1, a series of increasingly sophisticated Amateur satellites were launched under the OSCAR designation, which eventually became a globally recognized naming convention (OSCAR stands for Orbiting Satellite Carrying Amateur Radio) [5]. Each satellite bore a number, and later a country prefix (e.g., AO-7 for AMSAT OSCAR 7), indicating the progression and internationalization of the movement.

The Role of AMSAT

In 1969, the founding of AMSAT (Radio Amateur Satellite Corporation) in the United States [6] provided a formal structure to coordinate development, launch partnerships, and community engagement for Amateur satellite projects. AMSAT became the flagship organization behind many of the OSCAR missions, supporting collaborations with universities, space agencies, and international partners. Throughout the 1970s and 1980s, notable satellites such as AO-6, AO-7, and AO-10 introduced new capabilities: linear transponders for voice SSB communications, telemetry systems, and even early digital modes. AO-7, launched in 1974, is legendary for its longevity-it was presumed inactive for years until it miraculously resumed operation in 2002 and continues to function periodically to this day [6].

Global Expansion and Innovation

As the Amateur satellite community grew, so did the number of participating countries. Organizations such as AMSAT-DL (Germany), AMSAT-UK and others joined the international effort, contributing their own designs and missions. By the 1990s and 2000s, Amateur satellites were experimenting with digital voice, store-and-forward messaging systems, and advanced telemetry.

The Phase 3 series of satellites (e.g., AO-13 and AO-40) pushed boundaries further by entering highly elliptical orbits (HEO), allowing longer visibility windows and global communication coverage. These missions demonstrated the technical sophistication and ambition of the Amateur satellite service, offering real-world learning platforms for students and a proving ground for advanced RF and spacecraft systems.

The Emergence of CubeSats and the Modern Era

The 21st century brought a transformation with the advent of the CubeSat standard-a modular, low-cost satellite format adopted widely by universities and research institutes. Amateur radio payloads began flying on these small platforms, allowing for easier launch access and experimentation. Satellites such as AO-85 (Fox-1A), AO-91, and AO-92 featured FM repeaters, telemetry beacons, and scientific sensors, continuing the legacy of hands-on Amateur access to space.

However, the rapid growth of commercial space activity has brought new pressures on spectrum traditionally allocated for Amateur satellite use. Commercial operators, small satellite startups, and broadband constellations began to increasingly seek access to VHF, UHF, and microwave bands - some of which overlap or sit adjacent to Amateur allocations. This encroachment continues to pose challenges for the Amateur community, which must advocate vigorously to preserve its spectrum, demonstrate responsible and innovative use, and continue proving the scientific, educational, and public-benefit value of Amateur radio in space.

In recent years, Amateur satellite efforts have intersected with educational programs, space science, and even deep space exploration. The ARISS (Amateur Radio on the International Space Station) program allows students worldwide to speak directly with astronauts via Amateur radio - a powerful inspiration for future generations.

Geostationary and Lunar Aspirations

In addition to the successes and proliferation brought on by the advent of CubeSats and rideshare missions, the long-held aspiration of placing Amateur radio in geostationary (GEO) orbit became reality with QO-100, the first GEO satellite to host Amateur transponders [7]. Its fixed position and continuous (Eastern) coverage demonstrated the value of high-orbit Amateur assets, enabling stable long-distance communication and advanced digital experimentation. Although GEO opportunities remain rare and typically require hosted payload arrangements, QO-100 has renewed global interest in pursuing additional geostationary capabilities.

Amateur radio ambitions have also extended toward the Moon, with both proposed and realized missions. Notably, China's Chang'e program has carried amateur-compatible payloads, including the DSLWP-A1 and DSLWP-A2 microsatellites and Amateur radio hardware supporting lunar-orbit operations. These successes have proven that modest Amateur systems can function at lunar distances, inspiring new concepts for beacons and low-power transponders in cislunar space. Despite challenges like radiation, power limits, and high launch costs, modern miniaturization and rideshare opportunities continue to make future lunar Amateur radio projects increasingly realistic.

Legacy and Looking Forward

Today, over 100 OSCAR-designated satellites have been launched [5], and the tradition of Amateur satellites continues to evolve. From the Moon to Mars and low Earth orbit to geostationary experiments, Amateur operators are pushing boundaries not just in RF communications, but also in software-defined radios, autonomous spacecraft, and space-based experimentation. For countries like Canada, the Amateur satellite service offers a unique opportunity to contribute to global space innovation with relatively low barriers to entry. As organizations like AMSAT-CA take shape, they inherit a legacy of creativity, collaboration, and citizen science - one that has always treated space as a place for everyone.

Potential AMSAT Pursuits

As the rich history of the Amateur satellite activities and AMSAT organizations indicate, potential pursuits by AMSATs around the globe come in many forms, but they can be generally categorized as shown in **Tables 1 to 3**. Each of these categories have their own distinct advantages and disadvantages, resource needs, and timelines. They include space-based satellite initiatives, terrestrial and ground infrastructure initiatives, research, development, and technological innovation. Note that advocacy and policy efforts as well as education, outreach and youth programs are not considered explicitly here; AMSAT-CA believes that those are the joint responsibility of all AMSAT organizations and need to be incorporated to all technical activities to the greatest practicable extent.

Table 1: Summary of potential pursuits for space based satellite initiatives.

Category ID	Name	Description
SPC-1	Low Earth Orbit (LEO) Satellite Missions	Design, build, and launch small satellites (e.g. CubeSats) for LEO carrying Amateur radio payloads to maintain a diverse and sizable OSCAR fleet [8]. This includes FM voice repeater satellites and linear transponders to support two-way communications, and includes satellites by AMSAT organizations, as well as partnerships with universities to host Amateur payloads developed by AMSAT, such as the AMSAT US LT-1 package.
SPC-2	Highly Elliptical Orbit (HEO) Satellite Missions	Design, build, and launch small satellites to HEO with Amateur radio payloads or support such activities by others to take advantage of the highly favourable access provided by these types of orbits for two-way communication operations.
SPC-3	Geostationary (GEO) Satellite Missions	Pursue missions in GEO to provide wide-area, continuous coverage for Amateur radio communications. This could involve collaborating on hosted payloads similar or developing an AMSAT-CA GEO mission focused on North America to achieve persistent regional coverage.
SPC-4	Missions for the Lunar Surface and Cislunar Space	AMSAT organizations can support lunar surface and cislunar missions by developing Amateur radio payloads for orbiters, landers, and relays. Projects could include placing transponders or beacons on lunar missions for Earth-Moon communication experiments, supporting propagation studies, and enabling long-range DX via passive or active repeaters. In cislunar space,

		AMSATs can contribute to communications systems on space stations or vehicles orbiting the Moon or deep space CubeSats using Amateur frequencies, extending Amateur radio operations beyond Earth orbit.
SPC-5	Deep Space Satellite Missions	AMSAT organizations can pursue deep space missions by developing low-power Amateur radio payloads for interplanetary probes, flybys, or smallsat missions beyond cislunar space. Potential projects include beacon transmitters or data relays for receiving telemetry from deep space, supporting signal tracking, propagation studies, and long-range communications. Collaborations with space agencies or universities could enable Amateur access to spacecraft operating at Mars, Venus, or in heliocentric orbit, extending the reach and scientific utility of Amateur radio.
SPC-6	Amateur Radio Human Spaceflight Communications	Expand Amateur radio's presence in crewed space missions. This includes continued support for ARISS and partnering on new endeavors like placing Amateur radio stations on planned crewed government or commercial space stations (e.g. Vast Haven-1 [9], Voyager Starlab [10] etc). Such projects enable astronauts and space tourists to engage with the Amateur community (e.g. school contacts); increase the availability of ARISS-like voice contacts; and strengthen the permanent footprint of Amateur radio in space.
SPC-7	Amateur Satellite Constellations and Networks	Investigate the feasibility of coordinated Amateur satellite constellations for continuous coverage or new services. While individual OSCARs are often standalone initiatives, a strategic project could be to coordinate launches forming a constellation or mini-network that extends the reach and resilience of Amateur radio in space, and contribute to Amateur radio emergency communications on a global scale (e.g. one satellite always in range to relay digital communication).
SPC-8	Enabling Technologies for Amateur Satellite Missions	AMSAT organizations can develop enabling technologies (e.g. avionics) for space-based Amateur radio experiments. The possibilities are broad here, and only limited by available expertise and funds. If such technologies are

		released as “Open Source” projects after completion, the impact can be globally significant for the community.

Table 2: Summary of potential pursuits for terrestrial and ground infrastructure initiatives.

Category ID	Name	Description
GND-1	Global or Regional Amateur Ground Station Networks*	Develop and support a network of ground stations to track, and receive telemetry explicitly from Amateur satellites. For example, AMSATs can provide hardware-agnostic tools for pass predictions, telemetry decoding, and satellite status, and could integrate these with a broader network of volunteer Amateur ground stations.
GND-2	Command and Control Infrastructure for Amateur Satellites	Establish infrastructure programs to invest in robust, dedicated command stations and operation centers for Amateur satellites. Repurpose existing facilities following established international community precedent (e.g. Dwingeloo Radio Observatory, Netherlands [11]) or establish new ones. AMSATs can establish dedicated terrestrial control facilities or regional command posts to improve contact with Amateur satellites throughout their orbits, and serve as a “big gun” type station for Amateur satellite recovery, or in support of lunar and deep space operations.
GND-3	User Ground Equipment Development	Facilitate the development of affordable, easy-to-use ground equipment for use by new Amateurs. For example, an AMSAT might undertake projects to create reference designs for low-cost tracking antenna kits or software-defined radio (SDR) based satellite transceivers that individuals or schools can assemble, similar to the AMSAT US CubeSat Simulator [12]. This lowers the barrier to entry for satellite operation and encourages the uptake of Amateur satellite activities by newcomers.
GND-4	Internet Linked Amateur Radio Satellite Systems	Explore integration of Amateur satellite systems with terrestrial Internet, potentially via the use of commercial proliferated LEO (PLEO) constellations such as Starlink, or Telesat Lightspeed, following established precedent and practice from terrestrial projects such as Winlink and EchoLink.

GND-5	Satellite-based Emergency Communication	Develop plans and infrastructure for using Amateur satellites in emergency and disaster scenarios. Strategically, AMSATs can work with emergency communications groups to incorporate satellite capability into their plans, or undertake projects with real emergency support potential, such as internet linked systems, systems which provide near continuous or continuous coverage.
GND-6	High-Altitude Balloon (HAB) Projects	Initiate a HAB program that incorporates Amateur radio or contribute to existing HAB programs, for example the annual stratospheric balloon launches by the Canadian Space Agency (CSA) [13]. These projects serve as an accessible gateway to space: participants learn to build instruments, track flights via APRS/telemetry, and recover experiments, all while seeing practical applications of radio communication and data collection in the upper atmosphere. Projects can include creating a template for novices to follow for simple launches, or developing and launching more advanced HABs or payloads by an AMSAT organization.

*The development of global ground station networks with TX capabilities are deemed impractical for regulatory purposes; most countries, including Canada, do not allow satellite TT&C on the basis of an individual or club Amateur radio license. See more in Section 4.1 of CPC-2-6-01 [14].

Table 3: Summary of potential pursuits for research, development, and technological innovation.

Category ID	Name	Description
RDI-1	New-Generation Satellite Technologies	Invest in R&D for cutting-edge satellite communication technology to push the state of the art. Future projects can include developing SDR transponders to allow more flexible in-orbit reconfiguration, higher-speed digital telemetry downlinks, and advanced error-correcting protocols to improve signal quality. Experimenting with previously unused frequency bands (e.g. > 10 GHz microwave bands) for Amateur satellite use is another strategic pursuit to prevent encroachment by and spectrum loss to commercial interests, enabled by the proliferation of RF Monolithic Microwave Integrated Circuit (MMIC) technology developed commmercially for automotive radar applications.
RDI-2	Digital Communication and	Pioneer new digital modes and signal processing

	Signal Processing	techniques for Amateur satellites, for example LoRa, and DVB-S/S2/S2X. This can enable features like real-time video downlinks from spacecraft or efficient data exchange with minimalist stations, greatly enriching the Amateur satellite experience.
RDI-3	Satellite-Based Networking and Autonomy	Explore inter-satellite linking and autonomous constellation management for Amateur missions. Research projects might aim to have Amateur satellites communicate with each other or with ground relay servers to route messages (creating an ad-hoc space network). Such forward-looking research ensures that Amateur satellites remain at the forefront of innovation.
RDI-4	Citizen Science	Engage in novel research or space-based citizen science projects to leverage the Amateur community to support research relevant to the Amateur-satellite service by university-based groups or space agencies.

Comparative Analysis

Overview

To establish a clear technical vision for AMSAT-CA, and ensure that the finite resources of the organization are focused on what it deems its best interest; it is necessary to systematically evaluate concrete ideas among the categories for the potential pursuits presented in the previous section.

To assist with this, a weighted objectives chart (WOC) with a pairwise comparison matrix is used. This is a structured decision-making tool used to objectively evaluate multiple options (i.e. concrete AMSAT-CA project candidates) against a defined set of criteria. First, the relative importance of each criterion is determined by comparing them against one another in pairs, producing normalized weights that reflect organizational priorities. Each alternative is then scored against those weighted criteria, and the weighted scores are summed to produce an overall ranking. This method helps reduce subjective or emotional decision-making by making tradeoffs explicit, traceable, and repeatable. Its benefits include improved transparency, defensible justification for recommendations, consistent evaluation across many alternatives, and the ability to align technical, operational, financial, and programmatic considerations into a single quantitative framework.

This allows organized consideration of merits and demerits of each concrete AMSAT-CA project candidate, while incorporating the preferences of the Board of Directors and Membership via the definition of the criteria and the weighting, which can be updated from time to time, as this Strategic Plan evolves via annual reviews.

Evaluation Criteria

The evaluation criteria for the WOC is defined in **Table 4**.

Table 4: WOC evaluation criteria.

Criteria ID	Evaluation Criteria	Description	Weight (x/100)
C-1	Alignment with AMSAT-CA Mission	How well does the project support AMSAT-CA's core purpose: to promote, develop, and support Amateur radio in space?	10
C-2	Technical Feasibility	Is the project realistically achievable within current or near-future technological capabilities that AMSAT-CA can access?	30
C-3	Programmatic Feasibility	Is the project realistically achievable given the current or near-future resources (money,	40

		personnel, facilities) that AMSAT-CA can access?	
C-4	Degree of Proliferation	To what extent is the project proliferated in the AMSAT community? How much does it push boundaries in terms of technology, techniques, or applications? Does it allow AMSAT-CA to contribute something new to the state of the art?	10
C-5	Engagement and Identity Potential	How well does the project aid AMSAT-CA in defining its unique identity among the international AMSAT organizations? Does it provide a straightforward rally point for Canadian Amateur satellite enthusiasts?	10

The justification of criteria is as follows:

I. **Alignment with AMSAT-CA Mission:** Any projects undertaken by AMSAT-CA should be in clear alignment with the organization's mission, as defined in its Certificate of Incorporation, and reproduced earlier in this document.
II. **Technical Feasibility:** Projects need to be technically feasible, otherwise they are not a good use of AMSAT-CA's limited resources. Technical feasibility is demonstrated by a technical feasibility study, to be completed as part of project definition for any concrete technical project AMSAT-CA undertakes.
III. **Programmatic Feasibility:** Projects need to be feasible from the programmatic standpoint, otherwise they are doomed to fail. This includes access to money personnel, facilities, and compliance with the domestic and international regulatory framework. This is demonstrated by a program feasibility study, to be completed as part of project definition for any concrete technical project AMSAT-CA undertakes.
IV. **Degree of Proliferation:** Projects need to be novel in nature to facilitate AMSAT-CA developing its own identity; and not "claimed" by another AMSAT organization. Contributing to existing initiatives is great, but it should not be top priority for a new organization. The more proliferation, the lower the score.
V. **Engagement and Identity Potential:** The project needs to rally Canadian Amateur satellite enthusiasts to AMSAT-CA; establish AMSAT-CA's identity internationally, and they should lend itself to completion within a reasonable timeframe to demonstrate the organization's capability to undertake projects, and achieve success.

Weighted Objectives Chart

The Weighted Objectives Chart, due to its format and complexity, isn't suitable to be included in a document such as the Strategic Plan. It can be accessed at this link as a read-only spreadsheet on the AMSAT-CA website.

Discussion

Based on the WOC introduced in the previous section, the most suitable projects for AMSAT-CA are simple, one-way communication experiments deployed as hosted payloads on the lunar surface or in cislunar space. These efforts can be supported and enabled through the initial development of simple, open-source experiments (kits) for OSCAR operations and high-altitude balloons (HABs), helping bridge the gap between entry-level participation and the long-term lunar beacon objective. Together, these initiatives form a layered progression that allows both the organization and its contributors to build capability incrementally, without leaving novice participants behind.

Simple hosted payloads for the lunar surface or cislunar space are particularly well suited to AMSAT-CA. When restricted to one-way, beacon-style experiments, such projects become practical applications of existing and well-understood technologies, with most individual elements already demonstrated in prior missions. Similar to ARISS SSTV activities [15], these payloads also reinforce a broad and inclusive participation model: receiving a beacon requires neither an Amateur radio licence nor significant financial investment, while simultaneously enabling citizen science participation on a global scale.

This simplicity also keeps development costs, regulatory requirements, and facility needs within the capabilities of AMSAT-CA through a combination of grants and the organization's own resources. At the same time, the increasing cadence of Canadian and international lunar missions - including initiatives such as the Canadian Space Agency's participation in the ARTEMIS II lunar flyby [16], or the MINERAL mission by Magellan Aerospace [17] - creates realistic opportunities to secure hosted payload space. AMSAT-CA's close connections with key space industry participants further strengthen the feasibility of pursuing rideshare opportunities through programs such as the Canadian Space Agency's Lunar Exploration Accelerator Program (LEAP) [18].

Perhaps most importantly, projects of this kind remain rare, with only a small number of comparable examples such as LEV-1 and Chang'e 4. No AMSAT organization has yet established a strong international identity in this area. As a result, a lunar or cislunar beacon project provides AMSAT-CA with a compelling and distinctive objective around which Canadian Amateur satellite enthusiasts can rally, while also helping establish the organization's identity within the international AMSAT community. In addition, following the 2026 cancellation of the Lunar Gateway Space Station [19] and the Lunar Utility Rover [20], AMSAT-CA pursuing work in this domain would also represent a meaningful contribution to maintaining Canada's relevance within international lunar exploration efforts.

The development of affordable, open-source kits for OSCAR and HAB operations is equally important, as it provides a practical pathway for AMSAT-CA to build both individual and organizational capability toward the lunar beacon objective. These projects can engage not only experienced Amateur radio operators and space industry professionals, but also university students, high school students, and educators. For many newcomers, entry into the Amateur satellite service remains technically intimidating, and few affordable, turnkey solutions currently exist to lower this barrier. Developing accessible kits and educational resources therefore represents a significant opportunity for AMSAT-CA to broaden participation, foster technical growth, and cultivate long-term contributors. The impact of these efforts could be strengthened further through a mentorship program pairing end users with experienced AMSAT-CA members.

It is noted that the avoidance of GEO Amateur satellite work and constellations as a primary initiatives of AMSAT-CA are a strategic decision which indicates an organizational focus rather than exclusion. AMSAT-CA acknowledges that participation in or support to partner missions can provide valuable rehearsal and capability-building aligned with the strategic vision outlined in this plan. However, in the interest of organizational productivity and focus, AMSAT-CA's participation in these is subject to:

1. Formal evaluation by a dedicated Working Group established for this purpose, using the Opportunity Evaluation Sheet in **Appendix A** of this document.
2. Communication of the summary of the evaluation results by the distribution of the completed Opportunity Evaluation Sheet to the Board and the Membership via the dedicated e-mail lists maintained by the Secretary of AMSAT-CA.
3. Subsequent favorable ordinary resolution of the Membership **and** the Board, with the special resolution interpreted as defined in the most recent AMSAT-CA Bylaws [3].

The Strategic Vision

Our vision is to have AMSAT-CA be a globally recognized leader in lunar Amateur radio, as well as the empowerment of beginners in the Amateur-satellite service. To this end:

I. By 2030, design, test and publish open source designs for a simple HAB package, as well as an VHF/UHF OSCAR station, suitable for dissemination by beginners, especially university students, high school students, and educators.

II. By 2035, design, develop, and fly a simple, one-way, long term, low space, weight and power (SWaP) lunar experiment on a Canadian or international vehicle, designed for the broadest possible global audience.

Roadmap and Implementation

The implementation of the Strategic Plan will follow the roadmap outlined in this **Table 5**. Execution of the various stages will be the responsibility of several dedicated Working Groups established by the Board of Directors to support focused, high-output contributions while remaining fully aligned with the strategic plan.

The table also outlines the entry criteria and deliverables for each stage as well. It is noted that the various deliverables and the start of the subsequent stages are subject to the approval of the Board, following a formal report and presentation to the Board by a member of the Dedicated Working group responsible for execution, presentable at any meeting.

Execution of subsequent stages of the roadmap without formal board approval is not permitted, because it is deemed detrimental to AMSAT-CA's desire outlined earlier in this document to have the organization and the individual contributors build capability incrementally.

Table 5: High level roadmap for the implementation of the AMSAT-CA strategic vision.

Stage ID	Description	Entry	Deliverables	Dedicated Working Group Name
1	Open source affordable VHF/UHF OSCAR station design, development, testing, and public release	Establishment of working group	Proven station design is available on the AMSAT-CA Github with documentation	OSCAR Station Working Group
2	Open source HAB mission in-design, development, flight testing, and public release	Completion of Stage 1 Board	Completion of successful balloon flight campaign, supported by the Stage 1 station; and the proven balloon "bus" is available on the AMSAT-CA Github with documentation	Balloon Working Group
3	Lunar beacon concept study	Completion of Stage 2	Written concept and associated technical and program feasibility study approved by the board	Lunar Beacon Technical Working Group
4	Lunar beacon opportunity development	Completion of Stage 2	Resources secured for execution of the lunar project. Risk matrix and budget approved by the board	Lunar Beacon Program Support Working Group
5	Lunar beacon project	Completion of Stage 4	Completion of the lunar beacon project flow, according to the phase/gate review	Several, to be established at kick-off

			process outlined in the NASA Systems Engineering Handbook [21] and established at a later time.	

Staying Relevant

In order to make sure that this Strategic Plan remains relevant, it is subject to annual review. This is to be initiated by the President every year in May, six months after the elections of Directors and Officers takes place in November of each year, to ensure that new recruits have an opportunity to become familiar with AMSAT-CA before contributing to this document.

The annual review shall be conducted by the Working Group on Corporate Strategy, as elected by the Board at its discretion. The Working Group shall conduct an electronic member survey, and prepare an update, including a revised copy of this Strategic Plan for the annual general meeting for approval by vote of the members. The update of the Strategic Plan shall be done in a manner to avoid compromising existing initiatives which have been approved for execution by the BoD.

References

[1] "Memorandum of Understanding Between RAC and AMSAT-CA." Accessed May 16, 2026. URL: https://amsat-ca.org/assets/media/mou-rac-amsat-ca.pdf

[2] "AMSATs Around the World." Accessed May 16, 2026. URL: https://www.amsat.org/amsats-around-the-world/

[3] "AMSAT-CA Bylaws (August 18, 2025)." Accessed May 16, 2026. URL: https://amsat-ca.org/assets/media/amsat-ca-bylaws-agust-18-2025.pdf

[4] "OSCAR 1 Launched 50 Years Ago." Accessed May 16, 2026. URL: https://www.arrl.org/files/file/Technology/Bilsing.pdf

[5] "Orbiting Satellites Carrying Amateur Radio." Accessed May 16, 2026. URL: https://www.amsat.org/orbiting-satellites-carrying-amateur-radio/

[6] "AMSAT History." Accessed May 16, 2026. URL: https://www.amsat.org/amsat-history/

[7] "Es'hail-2 / QO-100." Accessed May 16, 2026. URL: https://amsat-uk.org/satellites/geo/eshail-2/

[8] "AMSAT Status Page." Accessed May 16, 2026. URL: https://www.amsat.org/status/

[9] "Haven-1." Accessed May 16, 2026. URL: https://www.vastspace.com/haven-1

[10] "Starlab." Accessed May 16, 2026. URL: https://voyagertechnologies.com/starlab/

[11] "CAMRAS." Accessed May 16, 2026. URL: https://www.camras.nl/en/

[12] "CubeSatSim Kit." Accessed May 16, 2026. URL: https://www.amsat.org/product/cubesatsim-kit/

[13] "STRATOS Program." Accessed May 16, 2026. URL: https://www.asc-csa.gc.ca/eng/sciences/balloons/stratos.asp

[14] "CPC-2-6-01 — Procedure for the Submission of Applications for Spectrum Licences and Site Approvals for Earth Stations Requiring Site Approval." Accessed May 16, 2026. URL: https://ised-isde.canada.ca/site/spectrum-management-telecommunications/en/learn-more/key-documents/procedures/client-procedures-circulars-cpc/cpc-2-6-01-procedure-submission-applications-license-fixed-earth-stations-and-approve-use-foreign

[15] "Upcoming SSTV Events." Accessed May 16, 2026. URL: https://www.ariss.org/upcoming-sstv-events.html

[16] "Artemis II Mission." Accessed May 16, 2026. URL: https://www.asc-csa.gc.ca/eng/missions/artemis-ii/mission.asp

[17] "Magellan Awarded Innovative New Lunar Program by Government of Canada." Accessed May 16, 2026. URL: https://magellan.aero/info-bite/magellan-awarded-innovative-new-lunar-program-by-government-of-canada/

[18] "Lunar Exploration Accelerator Program (LEAP)." Accessed May 16, 2026. URL: https://www.asc-csa.gc.ca/eng/funding-programs/programs/leap/

[19] "NASA Halts Work on Gateway to Develop a Lunar Base." Accessed May 16, 2026. URL: https://spacenews.com/nasa-halts-work-on-gateway-to-develop-a-lunar-base/

[20] "Canadian Space Agency Terminates Lunar Rover Mission in 2026–27 Plan." Accessed May 16, 2026. URL: https://spaceq.ca/canadian-space-agency-terminates-lunar-rover-mission-in-2026-27-plan/

[21] "NASA Systems Engineering Handbook." Accessed May 16, 2026. URL: https://www.nasa.gov/wp-content/uploads/2018/09/nasa_systems_engineering_handbook_0.pdf

Appendix A - Opportunity Evaluation Sheet

Introduction

Project Title	
Lead Organization Name	
Working Group Chair	
Date of Evaluation	

Project Overview

Estimated Start Date		Estimated Duration (months)	
Project Description			
Key Objectives			

Project Cost

Estimated Total Cost (CA$)	
AMSAT-CA Allocation from Total (CA$)	
Funding Provided by AMSAT-CA (CA$)	
Funding Sources for AMSAT-CA Funds	

Technical Resources

	State What Is Available	What What Is Missing
Personnel		
Facilities		
Equipment		

Organizational Bandwidth

Describe how this project will impact the implementation of the Strategic Plan.

Board Decision

Comments:	Decision:

End of Document